SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Technology Stack

Date	13 July 2026
Team ID	
Project Name	AgroSense AI – Intelligent Crop Recommendation system
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*HTML5, CSS3, JavaScript*	*Creates a responsive and user-friendly interface for entering soil parameters and displaying crop recommendations.*

2	**Backend / Server-Side**	*Python, Flask*	*Handles HTTP requests, validates user inputs, communicates with the machine learning model, and returns prediction results.*
3	**Database / Data Storage**	*CSV Dataset (Crop Recommendation Dataset), Joblib (.pkl Model)*	*The crop dataset is used to train the machine learning model. The trained model is stored as a serialized .pkl file and loaded during prediction. No relational database is used in the current project.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Vercel*	*Hosts the application online, enabling users to access the Crop Recommendation System through a web browser with simple deployment and reliable availability.*
5	**Version Control & CI/CD**	*Git, GitHub*	*Git is used for source code version control, while GitHub stores the repository and supports collaborative development and project management.*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn, NumPy, Pandas, Joblib*	*Scikit-learn is used for crop prediction, NumPy for numerical computations, Pandas for dataset processing, and Joblib for loading the trained machine learning model efficiently during prediction.*